CBET Memorization Sheet

Print this and keep it beside you. The CBET exam rewards recall of normal values, safety limits, device settings, units, standards, and formulas across every kind of equipment in a hospital. Every number on this sheet is fair game.

01 The exam itself

FACT	VALUE
Certifying body	AAMI Credentials Institute (ACI)
Questions	165 multiple choice; 150 scored, 15 unscored pretest items mixed in
Time and rules	3 hours, closed book, simple calculator allowed, no phones
Passing	Criterion-referenced cut score (modified Angoff), about 116 of 165 correct, roughly 70%
Healthcare Technology and Function	30%
Healthcare Technology Problem Solving	30%
Healthcare Information Technology	17%
Public Safety in the Healthcare Facility	10%
Anatomy and Physiology	7%
Fundamentals of Electricity and Electronics	6%

> Function and problem solving are 60% of the exam. For every device: what it measures or delivers, its settings and normal values, its alarms, its common faults, and the first check.

02 Anatomy and physiology

VITAL SIGN (ADULT, RESTING)	NORMAL RANGE	MEASURED BY
Heart rate	60 to 100 bpm (below 60 bradycardia, above 100 tachycardia)	ECG, pulse oximeter, NIBP
Respiratory rate	12 to 20 breaths per minute	Impedance pneumography, capnography
Blood pressure	About 120/80 mmHg; MAP = diastolic + (systolic minus diastolic) / 3, about 70 to 100	NIBP, arterial line
Oxygen saturation (SpO_2)	95 to 100%	Pulse oximeter
End-tidal CO_2 ($EtCO_2$)	35 to 45 mmHg	Capnograph
Body temperature	37 C (98.6 F); fever above about 38 C	Thermistor, thermocouple, infrared
Fetal heart rate	110 to 160 bpm	Doppler ultrasound
Tidal volume	About 500 mL at rest (6 to 8 mL/kg ideal body weight on a ventilator)	Ventilator, spirometer
Cardiac output	About 5 L/min = heart rate x stroke volume (about 70 mL)	Derived

SYSTEM	KEY PARTS	WHAT THE TECHNICIAN'S DEVICES DO WITH IT
Circulatory	Heart (right atrium, right ventricle, left atrium, left ventricle), tricuspid, pulmonary, mitral, aortic valves, arteries, veins, capillaries	ECG records electrical activity; NIBP and arterial lines measure pressure; defibrillators and pacemakers control rhythm; balloon pumps assist

SYSTEM	KEY PARTS	WHAT THE TECHNICIAN'S DEVICES DO WITH IT
Blood flow	Body, vena cava, RA, tricuspid, RV, pulmonary valve, pulmonary artery, lungs, pulmonary veins, LA, mitral, LV, aortic valve, aorta, body	Right side pumps to lungs, left side to body
Conduction path	SA node, AV node, bundle of His, bundle branches, Purkinje fibers	P wave atrial depolarization, QRS ventricular depolarization, T wave ventricular repolarization
Respiratory	Trachea, bronchi, bronchioles, alveoli, diaphragm	Ventilators move gas; oximeters measure oxygen carried; capnographs measure CO2 exhaled; spirometers measure volumes
Nervous	Brain (cerebrum, cerebellum, brainstem), spinal cord, peripheral and autonomic nerves	EEG records brain activity; EMG records muscle; stimulators test conduction; sympathetic raises heart rate, parasympathetic lowers it
Musculoskeletal	206 bones, joints, skeletal (voluntary), smooth (involuntary), and cardiac muscle	Physical therapy stimulators, traction, continuous passive motion
Gastrointestinal	Esophagus, stomach, small intestine (absorption), large intestine, liver, pancreas, gallbladder	Feeding pumps deliver enteral nutrition; suction removes; endoscopes view
Endocrine	Pancreas (insulin, glucagon), thyroid, adrenals (epinephrine), pituitary	Glucose meters, insulin pumps
Renal	Kidneys (nephrons filter blood), ureters, bladder	Dialysis replaces filtration
Skin	Epidermis, dermis, subcutaneous layer; barrier, temperature control, sensation	Electrode contact and skin impedance, burns, pressure injuries
Blood	Plasma, red cells (hemoglobin carries oxygen), white cells (immunity), platelets (clotting)	Analyzers count and measure; oximetry depends on hemoglobin

Directional terms and word parts

- Anterior front, posterior back; superior above, inferior below; medial toward the midline, lateral away; proximal near the trunk, distal far; supine face up, prone face down; planes: sagittal (left and right), coronal (front and back), transverse (top and bottom)
- Systole contraction, diastole relaxation. Hypertension high pressure, hypotension low. Hypoxia low oxygen, hypercapnia high CO2, apnea no breathing
- Prefixes: brady- slow, tachy- fast, hyper- high, hypo- low, a-/an- without, dys- difficult, peri- around, endo- inside
- Roots: cardio heart, pulmo/pneumo lung, nephro/reno kidney, hepat liver, neuro nerve, gastro stomach, derm skin, osteo bone, hemo/hemat blood, myo muscle, encephal brain
- Suffixes: -itis inflammation, -ectomy removal, -ostomy opening, -oscopy viewing, -graphy recording, -gram the record, -algia pain, -emia blood condition

ECG timing (normal adult)

- PR interval 0.12 to 0.20 s; QRS under 0.12 s; QT under about 0.44 s
- Paper speed 25 mm/s; 1 mV = 10 mm; small box 0.04 s (1 mm), large box 0.20 s (5 mm)
- Heart rate = 1500 / small boxes between R waves, or 300 / large boxes
- Ventricular fibrillation: chaotic, no QRS, shockable. Asystole: flat line, not shockable. Atrial fibrillation: irregular, no P waves, cardiovert synchronized

03 Electricity and electronics

FORMULA	STATEMENT
Ohm's law	V = I x R; I = V / R; R = V / I
Power	P = V x I = I squared x R = V squared / R (watts)

FORMULA	STATEMENT
Series resistance	R total = R1 + R2 + ... ; same current through each; voltages add
Parallel resistance	1 / R total = 1/R1 + 1/R2 + ... ; two resistors: (R1 x R2) / (R1 + R2); same voltage across each; currents add
Kirchhoff	Currents into a node equal currents out; voltages around a loop sum to zero
AC values	RMS = 0.707 x peak; peak = 1.414 x RMS; peak-to-peak = 2 x peak; US mains 120 V RMS at 60 Hz (many countries 230 V at 50 Hz)
Transformer	V secondary / V primary = N secondary / N primary; step-down lowers voltage and raises current; isolation transformer 1:1
Capacitive reactance	Xc = 1 / (2 x pi x f x C); capacitors pass AC, block DC
Inductive reactance	XL = 2 x pi x f x L; inductors pass DC, oppose changes in current
RC time constant	tau = R x C; capacitor reaches 63% of full charge in one tau, about 99% in five
Frequency and period	f = 1 / T; 60 Hz has a period of 16.7 ms
Stored energy	E = one half x C x V squared (joules); how a defibrillator capacitor stores its charge

- Resistor color code digits: black 0, brown 1, red 2, orange 3, yellow 4, green 5, blue 6, violet 7, gray 8, white 9; tolerance gold 5%, silver 10%
- US cord colors: black (or red) hot, white neutral, green or bare ground. IEC cords: brown live, blue neutral, green-yellow ground
- Schematic symbols: zigzag resistor, parallel lines capacitor, coil inductor, triangle-with-bar diode, ground, fuse, switch, battery; signals flow left to right; compare measured voltages to the values printed at test points
- Passive devices (resistors, capacitors, inductors, transformers) cannot add energy; active devices (diodes, transistors, ICs, op-amps) control or amplify
- Diode conducts one way, forward drop about 0.7 V silicon; zener regulates; LED lights; rectifiers: half-wave, full-wave, bridge; capacitor filters ripple; regulator holds output steady
- Linear power supply: heavy transformer, low noise, inefficient. Switching supply: light, efficient, more noise, needs a load
- Transistor: NPN or PNP, base controls collector-emitter; used as a switch or amplifier. SCR and triac switch AC power
- Op-amp rules: huge gain, inputs draw no current, feedback makes the inputs equal; instrumentation amplifier rejects common-mode noise (high CMRR) for ECG front ends; isolation amplifiers and optocouplers keep patient circuits floating
- Filters: low-pass keeps slow signals, high-pass removes baseline drift, band-pass keeps a range, notch removes 60 Hz
- DMM measures volts, amps (in series), ohms (power off); oscilloscope shows voltage against time, 10x probe divides by ten; function generator provides test signals

POWER STORAGE AND CONDITIONING	FACTS
Cell voltages	Lead-acid 2.0 V, nickel-cadmium 1.2 V, nickel-metal hydride 1.2 V, lithium-ion 3.6 to 3.7 V, alkaline 1.5 V
Battery behavior	NiCd memory effect (condition with full cycles); Li-ion no memory but thermal runaway risk and must not over-discharge; sealed lead-acid sulfates if left discharged; capacity in amp-hours falls with age
UPS types	Standby (switches on loss), line-interactive (regulates voltage), online double-conversion (always on inverter, best protection)
Power problems	Sag or brownout (low voltage), surge (high), transient spike, noise, blackout; conditioners regulate and filter
Hospital essential electrical system (NFPA 99 and NFPA 70)	Generator restores power within 10 seconds; branches: life safety, critical (patient care, red outlets), equipment; UPS bridges the gap for critical devices
Isolated power and LIM	Ungrounded secondary of an isolation transformer; line isolation monitor alarms at 5 mA total hazard current; used in wet procedure locations
GFCI	Trips at 4 to 6 mA of imbalance between hot and neutral in about 25 ms; wet locations

Electrical safety

CURRENT THROUGH THE BODY (60 HZ, HAND TO HAND)	EFFECT
1 mA	Threshold of perception (tingle)
5 mA	Accepted maximum harmless current
10 to 20 mA	Let-go threshold: muscles contract and the person cannot release
50 mA	Pain, possible fainting, respiratory arrest
100 to 300 mA	Ventricular fibrillation likely
6 A and above	Sustained heart contraction, burns, respiratory paralysis
Microshock	As little as 10 microamps delivered directly to the heart (through a catheter or pacing wire) can cause fibrillation

IEC 60601-1 LIMIT	NORMAL CONDITION	SINGLE FAULT CONDITION
Earth (ground) leakage	5 mA	10 mA
Touch (enclosure, chassis) current	100 microamps	500 microamps
Patient leakage, type B and BF, AC	100 microamps	500 microamps
Patient leakage, type CF, AC	10 microamps	50 microamps
Patient leakage, DC (all types)	10 microamps	50 microamps
Mains voltage on the applied part	BF: 5 mA; CF: 50 microamps	(single fault test)

NFPA 99 AND FACILITY FACTS	VALUE
Ground (chassis to plug ground pin) resistance	0.5 ohm maximum for patient-care equipment
Chassis (touch) leakage, cord-connected patient-care equipment	500 microamps in current editions (older editions used 300 or 100 microamps; know the limit your facility cites)
Patient care vicinity	About 6 feet (1.8 m) beyond the bed or chair and 7.5 feet (2.3 m) above the floor
Line isolation monitor alarm	5 mA total hazard current
GFCI trip	4 to 6 mA (nominal 5 mA)
Hospital-grade receptacle	Green dot; test retention force, polarity, ground continuity
Relocatable power taps	Only where allowed, rated, and secured; never daisy-chained; no extension cords for patient care equipment

Classes, types, symbols, and testing

- Class I: protective earth (three-prong ground). Class II: double insulation, no ground (square-in-square symbol). Internally powered: battery only
- Applied part type B: body, not for direct cardiac use (stick figure). BF: body floating, isolated (figure in a box). CF: cardiac floating, safe for direct heart connection (heart in a box)
- Defibrillation-proof applied part: the symbol with paddles; withstands a defibrillator discharge
- Macroshock: current through the skin; needs milliamps. Microshock: current bypassing the skin to the heart; microamps
- Safety analyzer sequence: visual inspection and cord check, ground resistance, chassis leakage (normal, reversed polarity, open ground, open neutral), patient lead leakage, lead-to-lead, mains on applied part
- Leakage rises with open ground and reversed polarity; a rising leakage trend on PM predicts insulation failure; test at incoming inspection, after repair, and per PM schedule

05 Infection control, hazards, fire, gas, radiation, MRI, and standards

Infection control

- Standard (universal) precautions: treat all blood and body fluids as infectious; hand hygiene before and after every patient contact and before touching equipment
- Transmission-based precautions: contact (gown, gloves), droplet (surgical mask), airborne (N95 respirator, negative-pressure room)
- PPE donning: gown, mask, eye protection, gloves. Doffing: gloves, eye protection, gown, mask (clean hands between)
- Cleaning removes soil; disinfection kills most microbes; sterilization kills all including spores
- Spaulding: critical items (enter sterile tissue) sterilize; semi-critical (touch mucous membranes) high-level disinfect; non-critical (intact skin) low-level disinfect
- Steam autoclave: 121 C (250 F) at 15 psi for 15 to 30 minutes, or 132 to 135 C flash cycles; biological indicators verify; ethylene oxide and hydrogen peroxide plasma for heat-sensitive items
- Clean and disinfect equipment before servicing it; sharps in puncture-resistant containers; OSHA bloodborne pathogens standard covers exposure control; report needlesticks immediately

Hazard communication and signage

- Safety data sheet (SDS): 16 sections including identification, hazards, first aid, fire fighting, handling and storage, exposure controls and PPE; must be accessible to staff
- GHS pictograms: flame (flammable), flame over circle (oxidizer), skull (acute toxicity), corrosion, exclamation mark (irritant), health hazard (silhouette), gas cylinder, environment
- NFPA 704 diamond: blue health, red flammability, yellow reactivity, white special (W with a line means water reactive, OX oxidizer); scale 0 to 4
- Laser signs state the class and wavelength; classes 1 (safe) through 4 (fire and eye hazard); wavelength-specific eyewear
- Ionizing radiation: trefoil symbol; time, distance, shielding; ALARA; inverse square law; lead aprons about 0.5 mm; dosimeters
- MRI: zones I (public) through IV (magnet room); the magnet is always on; ferromagnetic objects become projectiles; quench vents helium; MR Safe, MR Conditional, MR Unsafe labels
- Patient precaution signs at the door: contact, droplet, airborne, fall risk, latex allergy, no blood pressure on a limb

Fire and medical gas

- Fire triangle: heat, fuel, oxygen. RACE: rescue, alarm, confine, extinguish. PASS: pull, aim, squeeze, sweep
- Extinguisher classes: A ordinary combustibles, B flammable liquids, C electrical, D metals, K cooking oils
- Oxygen-enriched atmospheres burn faster; no oil or grease on oxygen fittings; cylinders secured upright, capped, stored below 125 F, full separated from empty
- US medical gas cylinder colors: oxygen green, nitrous oxide blue, carbon dioxide gray, nitrogen black, medical air yellow, helium brown; pin index and DISS fittings prevent wrong connections
- Full E cylinder of oxygen: about 2,000 to 2,200 psi, about 660 liters; wall outlets about 50 psi; zone valves shut off gas to an area

STANDARDS, REGULATORS, AND ACCREDITATION	WHAT TO KNOW
CMS	Centers for Medicare and Medicaid Services; sets Conditions of Participation hospitals must meet to be paid; can survey directly
Deemed status	Accrediting organizations survey on CMS's behalf: The Joint Commission, DNV Healthcare, ACHC (formerly HFAP), CIHQ
The Joint Commission	Environment of Care and Emergency Management standards cover equipment inventory, maintenance strategies, testing, and documentation
FDA	Regulates devices; medical device reporting: user facilities report a device-related death to FDA and the manufacturer within 10 work days, a serious injury to the manufacturer within 10 work days; recalls Class I (serious harm), II (temporary harm), III (unlikely harm)
OSHA	Worker safety: hazard communication, bloodborne pathogens, PPE, electrical safety in the workplace
ANSI/AAMI EQ56	Recommended practice for a medical equipment management program
ANSI/AAMI EQ89	Guidance for scheduled maintenance and performance verification

STANDARDS, REGULATORS, AND ACCREDITATION	WHAT TO KNOW
ANSI/AAMI EQ93	Medical equipment management vocabulary and terminology
ANSI/AAMI EQ103	Alternative equipment maintenance (AEM) programs: how to justify and document maintenance strategies that differ from the manufacturer's
AEM rules (CMS)	Allowed with documented risk assessment by qualified staff; not for imaging or radiologic equipment, lasers, or where the manufacturer or law requires otherwise; new equipment follows the manufacturer until history supports a change
CMMS	Computerized maintenance management system: inventory and asset tags, work orders, PM schedules, equipment history, parts, reports (PM completion rate, mean time between failures, cost); every activity documented there
After an incident	Remove the device with all accessories and disposables attached, do not clear logs or settings, tag and sequester, document, report; root cause analysis asks why repeatedly

06 Physiological monitoring

ECG FACT	VALUE
Einthoven's triangle	Lead I = LA minus RA; Lead II = LL minus RA; Lead III = LL minus LA; Einthoven's law: Lead II = Lead I + Lead III
Augmented and chest leads	aVR, aVL, aVF from the limb electrodes; V1 to V6 across the chest; 12 leads from 10 electrodes
Electrode colors (US, AHA)	RA white, LA black, RL green (ground), LL red, V brown ("white on right, smoke over fire")
Bandwidth	Monitoring mode 0.5 to 40 Hz; diagnostic mode 0.05 to 150 Hz
Interference	60 Hz: bad electrode contact, unshielded leads, nearby equipment; wandering baseline: respiration or motion; spikes: muscle tremor or loose lead
Lead-off	High impedance detected on an electrode; check gel, skin prep, cable
Simulator test	Patient simulator provides rate, amplitude, and arrhythmias to verify the monitor

BLOOD PRESSURE FACT	VALUE
Korotkoff sounds	Auscultatory method: cuff above systolic, deflate; first sound (phase 1) is systolic, sounds muffle then disappear (phase 5) at diastolic; automated devices use oscillometry instead
NIBP method	Oscillometric: cuff inflates above systolic, deflates in steps, MAP is the point of maximum oscillation, systolic and diastolic derived
Cuff sizing	Bladder width about 40% of arm circumference, length about 80%; too small reads high, too large reads low
Invasive line	Transducer zeroed to atmosphere and leveled at the phlebostatic axis (fourth intercostal space, midaxillary line)
Dicrotic notch	The small dip on the downslope of the arterial waveform marking aortic valve closure; its absence suggests overdamping
Height error	13.6 cm of water equals 10 mmHg; every 13.6 cm the transducer is off level changes the reading about 10 mmHg
Damping	Fast-flush (square wave) test: overdamped rounds the wave and reads low systolic; underdamped overshoots and reads high
Transducer sensitivity	5 microvolts per volt per mmHg (industry standard)

OXIMETRY AND CAPNOGRAPHY	VALUE
Pulse oximeter wavelengths	Red about 660 nm, infrared about 940 nm; ratio of absorbances gives SpO2
Errors	Carboxyhemoglobin reads falsely high; methemoglobin trends toward 85%; motion, low perfusion, nail polish, ambient light, dyes
Testing	SpO2 simulator or optical tester; check probe LED and detector, cable, and site
Capnography	Infrared absorption by CO2 (about 4.26 micrometers); mainstream sensor at the airway, sidestream samples through tubing (delay, water trap)
Capnogram	Baseline near zero, rapid rise, alveolar plateau, EtCO2 at end of plateau, drop on inspiration; loss of waveform means disconnect, apnea, or esophageal intubation

Other monitoring

- Temperature: thermistor (resistance falls as temperature rises, NTC), thermocouple (voltage from two metals), infrared tympanic; YSI 400 and 700 series probe curves are not interchangeable
- Respiration on a monitor: impedance pneumography through the ECG electrodes; capnography is more reliable
- Alarm priorities (IEC 60601-1-8): high red flashing, medium yellow flashing, low cyan or yellow steady; alarm fatigue is a patient safety problem; alarm management programs set limits per unit
- Telemetry: WMTS bands 608 to 614 MHz, 1395 to 1400 MHz, 1427 to 1432 MHz; dropouts from antenna coverage, low battery, interference; central station shows all beds
- Fetal monitor: ultrasound Doppler for fetal heart rate (110 to 160 bpm), tocodynamometer (TOCO) for contractions by pressure on the abdomen; internal scalp electrode and intrauterine pressure catheter are the direct methods
- EEG bands: delta under 4 Hz, theta 4 to 8, alpha 8 to 13, beta above 13; electrodes placed by the 10-20 system; EMG measures muscle electrical activity; evoked potentials time the response to a stimulus

07 Test equipment and units of measure

TEST EQUIPMENT	VERIFIES
Electrical safety analyzer	Ground resistance, chassis and patient lead leakage under normal and fault conditions
Patient (physiological) simulator	ECG rate, amplitude, arrhythmias, respiration, invasive pressure, temperature
SpO2 simulator	Saturation and pulse readings through the probe
NIBP simulator	Cuff pressures and oscillometric readings; leak test
Defibrillator analyzer	Delivered energy into 50 ohms, charge time, synchronization, pacing output
ESU analyzer	Output power into a load, leakage, return electrode monitoring
Gas flow and ventilator analyzer	Volume, flow, pressure, oxygen concentration
Infusion pump analyzer	Flow rate accuracy, occlusion pressure
DVM or DMM and meters	Voltage, current, resistance; oxygen analyzer, pressure meter, tachometer, light meter, sound level meter
Cable tracer or toner	Finds the far end of a network cable

UNIT	CONVERSION OR MEANING
mmHg	Blood and gas pressure; 760 mmHg = 1 atmosphere; 1 mmHg = 1.36 cm H2O
cm H2O	Ventilator and airway pressures; 1 cm H2O = 0.74 mmHg
psi	Gas cylinders and pipelines; 14.7 psi = 1 atmosphere; 1 psi = 51.7 mmHg; 1 psi = 6.9 kPa

UNIT	CONVERSION OR MEANING
bar	1 bar = 100 kPa = 14.5 psi = 750 mmHg, about 1 atmosphere
kPa	1 kPa = 7.5 mmHg; 101.3 kPa = 1 atmosphere
Fahrenheit and Celsius	F = C x 1.8 + 32; C = (F minus 32) / 1.8; 37 C = 98.6 F; 0 C = 32 F; 100 C = 212 F
Joules	Energy: 1 J = 1 watt-second; defibrillator output
Lumens and lux	Lumens: total light output of a lamp; lux: lumens per square meter at the surface (surgical lights are rated in lux at 1 m)
Decibels	Sound level and audiometer hearing threshold (dB HL); 10 dB is a tenfold power ratio
Weight and length	1 kg = 2.2 lb; 1 inch = 2.54 cm; 1 L = 1000 mL; 1 mL = 1 cc
Prefixes	kilo 10 to the 3, milli 10 to the minus 3, micro 10 to the minus 6, nano 10 to the minus 9

08 Diagnostic, infusion, therapeutic, and laboratory equipment

DIAGNOSTIC DEVICE	FACTS
Otoscope, ophthalmoscope	Battery or wall-powered light and optics for ear and eye; check lamp, battery, charging contacts, specula
Audiometer	Hearing thresholds in dB HL at 250 to 8000 Hz; calibrate with a sound level meter and coupler; quiet room or booth
Spirometer	FVC, FEV1, FEV1/FVC ratio (normal about 0.7 to 0.8); calibrate with a 3 L syringe; filters and mouthpieces
Stress test system	Treadmill or bike with 12-lead ECG and NIBP; emergency stop; treadmill speed and grade calibration; defibrillator nearby
Scales	Calibrate with certified test weights; zero with the bed or chair empty; load cells
Ultrasound	Piezoelectric transducer sends and receives; 2 to 15 MHz; higher frequency, better resolution, less penetration; Doppler measures flow; gel removes air; transducer crystals crack if dropped

INFUSION DEVICE	FACTS
Large-volume pump	Peristaltic; accuracy typically within 5%; rate in mL/hr; free-flow protection; occlusion and air-in-line alarms; drug library
Syringe pump	Precise small volumes; syringe size must be recognized; start-up delay and siphoning risks
PCA pump	Patient button, bolus dose, lockout interval, hourly limit; locked to prevent tampering
Feeding (enteral) pump	Enteral-only connectors on purpose; lower accuracy acceptable
Contrast injector	Programmed volume and flow for imaging; pressure limit; air detection; heated syringe
Rate math	mL/hr = total mL / hours; drops per minute = (mL/hr x drop factor) / 60; mL/hr = (mg/kg/hr x kg) / (mg/mL)

THERAPEUTIC DEVICE	FACTS
Infant incubator and warmer	Air and skin temperature servo control, humidity, oxygen; radiant warmer with skin probe; over-temperature alarms; phototherapy blue light about 425 to 475 nm
Patient temperature management	Hypo- and hyperthermia units circulate water through blankets or pads; water temperature limits about 4 to 42 C; check flow, leaks, temperature sensor
Aspiration (suction)	Regulators in mmHg, continuous or intermittent; adult range about 80 to 120 mmHg; check filter, collection canister, tubing, vacuum source
Sequential compression device	Inflates leg sleeves in sequence to prevent clots; check pressures and cycle timing, hoses, sleeves
Physical therapy equipment	TENS and muscle stimulators, therapeutic ultrasound (1 to 3 MHz), diathermy, traction, continuous passive motion; output and timer verification

LABORATORY DEVICE	FACTS
Centrifuge	Speed in RPM, force in RCF (g); balance opposite loads; lid interlock; verify speed with a tachometer and timer
Incubator	Temperature (often 37 C) and sometimes CO2 and humidity; verify with a reference thermometer; door seals
Rocker and shaker	Speed and tilt; motor and belt
Refrigerator and freezer	Blood and vaccine storage 2 to 8 C; freezers minus 20 or minus 80 C; continuous temperature logging and alarms; door gaskets and defrost
Microscope	Light source, objectives, focus; cleaning optics
Water bath	Temperature control and uniformity; thermostat and heater
Analyzers	Reagents, calibration, quality control samples, fluidics and clogs, temperature
Cryostat and microtome	Cryostat cuts frozen sections in a cold chamber (about minus 20 C); microtome cuts thin sections; blade safety, temperature, advance mechanism

09 Perioperative and life-support devices

ELECTROSURGERY FACT	VALUE
Frequency	About 300 kHz to 3 MHz, above the frequency that stimulates nerves and muscle
Modes	Cut: continuous sine wave, vaporizes. Coag: interrupted or damped bursts, dries and seals. Blend mixes them
Monopolar	Current flows from the active electrode through the patient to a large return (dispersive) electrode; return electrode contact quality monitoring prevents burns
Bipolar	Current flows between the two tips of the forceps; no return pad needed
Burns	Poor return pad contact, pad on bony or hairy site, alternate paths through ECG electrodes or metal; capacitive coupling in laparoscopy
Video integration	Cameras, light sources, monitors, recording and routing systems; check cable, connector, light cable fibers, white balance

Other perioperative equipment

- Pneumatic tourniquet: cuff pressure above systolic (commonly 250 mmHg arm, 300 mmHg leg or adjusted to the patient), time limit alarms (about 2 hours), leak test, calibrated gauge
- Fluid and blood warmer: keeps fluids near body temperature; over-temperature cutoff around 41 to 42 C; check heater, sensor, alarms
- Sterilizer: steam 121 C at 15 psi or 132 C flash; verify temperature, pressure, time, gasket, chamber; biological indicator; ethylene oxide and hydrogen peroxide for heat-sensitive items
- OR table: hydraulics or motors, controls, brakes, weight rating; surgical lights: lux at 1 m, color temperature, handles, bulbs or LEDs; surgical microscope: optics, illumination, balance, focus motors

DEFIBRILLATION AND PACING	VALUE
Adult energy	Monophasic 360 J; biphasic 120 to 200 J (device specific); pediatric 2 J/kg first, 4 J/kg after
Synchronized cardioversion	Shock timed to the R wave for organized rhythms (atrial fibrillation, SVT); unsynchronized for VF and pulseless VT
Testing	Defibrillator analyzer with a 50 ohm load; check delivered energy, charge time, synchronization, pacing output; daily user checks
AED	Analyzes rhythm, advises shock only for VF and pulseless VT

DEFIBRILLATION AND PACING	VALUE
External pacing	Demand mode paces only when intrinsic rate falls below the set rate; fixed mode paces regardless; output in mA; capture confirmed by QRS after each spike
Intra-aortic balloon pump	Balloon in the descending aorta inflates in diastole (raises coronary perfusion) and deflates just before systole (lowers afterload); timed to the ECG or arterial waveform; helium gas; alarms for timing, leak, and gas loss

VENTILATOR AND ANESTHESIA	VALUE
Tidal volume and minute volume	6 to 8 mL/kg ideal body weight; minute volume = tidal volume x rate
Typical settings	Rate 12 to 20; I:E ratio 1:2; PEEP 5 cm H_2O; FiO_2 0.21 to 1.0
Modes	Volume control delivers a set volume; pressure control a set pressure; assist-control, SIMV, pressure support, CPAP; BiPAP has separate inspiratory and expiratory pressures
Alarms	High pressure: obstruction, kink, coughing, secretions. Low pressure or low volume: disconnect or leak. Apnea: no breath detected
Oxygen analyzer calibration	Room air 21% and 100% oxygen; galvanic cells wear out; concentrators deliver about 90 to 95%
Anesthesia machine	Pipeline (50 psi) and cylinder supply, flowmeters, agent-specific vaporizers (sevoflurane yellow, isoflurane purple, desflurane blue), hypoxic guard keeps oxygen at least 21 to 25%, oxygen flush, CO_2 absorber, scavenging removes waste gas, pin index and DISS prevent misconnection; leak test before use

10 Problem solving and troubleshooting

The method, in order

- 1. Confirm the complaint: talk to the user, reproduce the problem, rule out use error (wrong settings, wrong accessory, wrong technique), and check the right disposables.
- 2. Check the simple things: power, plug, battery, fuse, cables, connectors, settings, sensors.
- 3. Isolate: which subsystem (power, sensor, signal path, processor, display, mechanical); half-split the path with a simulator and test equipment; localized device or system-wide?
- 4. Form and test one hypothesis at a time; swap known-good parts; use the schematic.
- 5. Repair or replace; follow the service manual.
- 6. Verify: full performance test and electrical safety test after any repair, before return to service.
- 7. Document in the CMMS: symptom, cause, action, parts, test results, time. Educate the user if it was a use error.

SYMPTOM	FIRST SUSPECTS
Device dead, no power	Outlet, cord, plug, fuse, power switch, battery, power supply output
Runs on AC but not battery	Battery aged or not charging; charger circuit; battery connector
Intermittent reboot or errors	Failing power supply capacitors, loose connector, overheating
ECG shows 60 Hz noise	Electrode contact, gel, skin prep, lead cable shield, nearby equipment, ground
Wandering baseline	Patient movement or respiration, loose electrode, high-pass filter setting
Intermittent signal	Cable flex failure near connectors, loose connector, cracked solder
NIBP reads high	Cuff too small, cuff too low, patient movement, hose leak causing slow deflation
Arterial line reads low and rounded	Overdamped: air bubble, clot, kinked tubing, loose connection

SYMPTOM	FIRST SUSPECTS
SpO2 no reading	Probe placement, perfusion, ambient light, probe LED failure, cable
Ventilator high-pressure alarm	Kinked circuit, secretions, patient coughing, water in tubing
Ventilator low-pressure or volume alarm	Disconnect, leak, cuff leak, circuit crack
Infusion pump occlusion alarm	Kinked tubing, closed clamp, clotted line, set loaded wrong
Infusion pump air-in-line alarm	Air in set, sensor dirty, set not seated
Defibrillator low delivered energy	Capacitor aging, relay, paddles or pads, battery
ESU patient burn	Return electrode contact, alternate path, damaged cable
Telemetry dropouts on one patient	Transmitter battery, leads, distance; many patients: antenna system or receiver, a system-wide problem
Monitor not sending to the central station or EMR	One bed: cable, port, network config on that monitor. Every bed: switch, server, middleware, network outage
Centrifuge vibration	Unbalanced load, worn rotor or bearing
Refrigerator temperature alarm	Door seal, defrost cycle, compressor, probe, overloading

Priorities, communication, maintenance

- Priority order: life support and high-risk devices first, then patient risk, then urgency and downtime and clinical impact; loaners for critical downtime
- Device error versus use error: a device that performs to specification but was used wrongly needs education and possibly a design or labeling report, not a repair
- In-service and cross-training: teach proper use, care, cleaning, and alarm response; document attendance; communicate clearly with clinicians, staff, and manufacturers in writing and verbally
- Preventive maintenance: scheduled or risk-based; AEM programs allowed with documented justification (not imaging, lasers, or where the manufacturer requires); PM after repair; performance verification against tolerances
- Calibration traceable to national standards; test equipment has its own calibration schedule
- Incoming inspection before first use: physical, functional, safety test, inventory entry, manuals on file; lifecycle ends with data wiped, decontamination, and disposition

11 Healthcare information technology

NETWORKING FACT	VALUE
IPv4 address	Four numbers 0 to 255; subnet mask separates network and host; default gateway is the exit router; same subnet required to talk without a router
MAC address	48-bit hardware address in 12 hex digits, burned into the network card; switches forward by MAC
Private ranges	10.0.0.0/8; 172.16.0.0 to 172.31.255.255; 192.168.0.0/16
DHCP and static	DHCP assigns addresses automatically; medical devices often use static addresses or DHCP reservations so they are always reachable
DNS	Names to addresses; ping by address works but by name fails means DNS
APIPA	169.254.x.x means DHCP failed
Ports	HTTP 80, HTTPS 443, SSH 22, DICOM commonly 104 or 11112, HL7 commonly 2575 (site specific)
Devices	Switch connects devices in a LAN by MAC; router connects networks by IP; access point connects Wi-Fi clients; access point controller manages many APs; KVM switch shares one keyboard, video, and mouse among servers; virtual servers run many operating systems on one host; LAN is local, WAN spans sites, cloud is provider-hosted

NETWORKING FACT	VALUE
Cabling	Cat 5e 1 Gbps, Cat 6A 10 Gbps, 100 m limit; fiber for longer runs; link light off means Layer 1; cable tracer finds the far end
Wi-Fi	2.4 GHz longer range, channels 1, 6, 11; 5 GHz more channels, less interference; WPA2 or WPA3; enterprise mode uses 802.1X with RADIUS
Telemetry and RTLS	WMTS bands for patient telemetry; real-time location systems track equipment with RF, infrared, or ultrasound tags

COMMAND OR TOOL	USE
ping	Is the address reachable, and how fast; ping the gateway first
ipconfig (Windows), ifconfig or ip (Linux)	Show the device's address, mask, gateway; release and renew DHCP
tracert or traceroute	Every hop to a destination; where the path breaks
nslookup	Test name resolution
arp -a	IP to MAC table; duplicates
netstat	Open connections and listening ports
Cable tracer and tester	Find the far end; check pinout and continuity

INTEROPERABILITY	MEANING
HL7 version 2	Text messages between systems: ADT (admit, discharge, transfer), ORM (orders), ORU (results, including device data)
HL7 FHIR	Modern web-based standard using resources and APIs
DICOM	Standard for medical images and their transfer; modality worklist pulls patient data to the device; PACS stores and serves images
EMR and EHR	Electronic medical record (within one organization) and electronic health record (across); devices feed vitals in directly or through middleware
Middleware	Translates and routes between devices and the EMR; device gateways; where most integration faults are diagnosed
Time synchronization	NTP keeps device clocks aligned so records and alarms line up

Protected data and cybersecurity

- HIPAA: privacy rule (who may see PHI), security rule (administrative, physical, technical safeguards), breach notification; PHI is any identifiable health information; minimum necessary
- NIST Cybersecurity Framework functions: govern, identify, protect, detect, respond, recover
- HITRUST: a certifiable framework combining many requirements; HHS cybersecurity performance goals: essential and enhanced practices for hospitals (MFA, patching, backups, segmentation, training)
- MDS2: manufacturer disclosure statement for medical device security; SBOM: software bill of materials
- Controls: unique accounts and MFA, least privilege, encryption at rest and in transit, patching on the vendor's validated schedule, network segmentation for devices that cannot be patched, antivirus where the vendor allows, physical security of workstations, screen locks
- Backups: 3 copies, 2 media, 1 offsite; test restores; ransomware response: isolate, report, restore
- Incident: disconnect the device from the network, do not power off if forensics are needed, report to IT security and the vendor, document

Computer hardware and software troubleshooting

- Hardware: power supply (no power, random reboots), hard drive (slow, errors, clicking; SMART warnings; replace and restore image), memory (crashes), peripherals (drivers, cables, ports), connectors and cables (reseat, swap)
- Software: Windows, Office, Linux, and Unix updates and patches only on the vendor's validated schedule for medical devices; drivers after updates; event logs; restore from image; user accounts and permissions
- Order of checks: power and cables, link lights, address and gateway, ping the gateway, ping the server, name resolution, application or service, then escalate to IT or the vendor